

Software Requirements Specification (SRS)

AI-Based Smart HVAC Controller

1. Introduction

The AI-Based Smart HVAC Controller is a MATLAB/Simulink and Stateflow-based project that uses machine learning to determine HVAC operating modes based on cabin temperature, setpoint temperature, humidity, solar load, and ambient temperature.

2. Scope

The system provides intelligent HVAC mode selection including Idle, Cooling, High Cooling, Heating, and Defog modes. The project demonstrates Model-Based Design, machine learning integration, Stateflow control logic, and MIL verification.

3. Functional Requirements

FR_001: HVAC Enable/Disable Control
FR_002: Cooling Mode Selection
FR_003: High Cooling Mode Selection
FR_004: Heating Mode Selection
FR_005: Defog Mode Selection
FR_006: AI HVAC State Prediction
FR_007: Temperature Error Calculation
FR_008: Humidity Monitoring
FR_009: Solar Compensation Processing

4. Non-Functional Requirements

- Modular Simulink Architecture
- Stateflow-based Control Logic
- Machine Learning Accuracy Target >90%
- Maintainable and Reusable Design
- Support for MIL Testing

5. Inputs

- T_Cabin
- T_Setpoint
- Humidity Sensor
- Q_Solar
- T_Ambient
- HVAC_Enable
- HVAC_MODE_REQ

6. Outputs

- Heater_Command
- Compressor_Command
- Blower_Speed
- Recirc_Command
- Airflow_Mode
- AI_HVAC_State

7. Operating Modes

State 1: IDLE

State 2: COOLING

State 3: HIGH_COOLING

State 4: HEATING

State 5: DEFOG

8. Assumptions

- Sensor inputs are valid.
- HVAC hardware responds correctly.
- AI model is trained with representative data.
- Simulation environment is correctly configured.

9. Constraints

- MATLAB R2016b compatibility.
- Limited training dataset size.
- MIL testing only in current phase.
- AI prediction limited to trained operating conditions.